```
            sepc=0x49be stval=0x28013000
usertrap(): unexpected scause 0xf pid=7832
            sepc=0x49be stval=0x29013000
usertrap(): unexpected scause 0xf pid=7833
            sepc=0x49be stval=0x2a013000
usertrap(): unexpected scause 0xf pid=7834
            sepc=0x49be stval=0x2b013000
usertrap(): unexpected scause 0xf pid=7835
            sepc=0x49be stval=0x2c013000
usertrap(): unexpected scause 0xf pid=7836
            sepc=0x49be stval=0x2d013000
usertrap(): unexpected scause 0xf pid=7837
            sepc=0x49be stval=0x2e013000
usertrap(): unexpected scause 0xf pid=7838
            sepc=0x49be stval=0x2f013000
usertrap(): unexpected scause 0xf pid=7839
            sepc=0x49be stval=0x30013000
usertrap(): unexpected scause 0xf pid=7840
            sepc=0x49be stval=0x31013000
usertrap(): unexpected scause 0xf pid=7841
            sepc=0x49be stval=0x32013000
usertrap(): unexpected scause 0xf pid=7842
            sepc=0x49be stval=0x33013000
usertrap(): unexpected scause 0xf pid=7843
            sepc=0x49be stval=0x34013000
usertrap(): unexpected scause 0xf pid=7844
            sepc=0x49be stval=0x35013000
usertrap(): unexpected scause 0xf pid=7845
            sepc=0x49be stval=0x36013000
usertrap(): unexpected scause 0xf pid=7846
            sepc=0x49be stval=0x37013000
usertrap(): unexpected scause 0xf pid=7847
            sepc=0x49be stval=0x38013000
usertrap(): unexpected scause 0xf pid=7848
            sepc=0x49be stval=0x39013000
usertrap(): unexpected scause 0xf pid=7849
            sepc=0x49be stval=0x3a013000
usertrap(): unexpected scause 0xf pid=7850
            sepc=0x49be stval=0x3b013000
usertrap(): unexpected scause 0xf pid=7851
            sepc=0x49be stval=0x3c013000
usertrap(): unexpected scause 0xf pid=7852
            sepc=0x49be stval=0x3d013000
usertrap(): unexpected scause 0xf pid=7853
            sepc=0x49be stval=0x3e013000
usertrap(): unexpected scause 0xf pid=7854
            sepc=0x49be stval=0x3f013000
OK
test lazy_copy: OK
test lazy_copyinstr: OK
test lazy_sbrk: OK
test partial_write: OK
test unlinkcwd: OK
ALL TESTS PASSED
$
```

```
riscv64-unknown-elf-objdump -t user/_zombie | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/zombie.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/logstress.o user/logstress.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_logstress user/logstress.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_logstress > user/logstress.asm
riscv64-unknown-elf-objdump -t user/_logstress | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/logstress.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/forphan.o user/forphan.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_forphan user/forphan.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_forphan > user/forphan.asm
riscv64-unknown-elf-objdump -t user/_forphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/forphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/dorphan.o user/dorphan.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_dorphan user/dorphan.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.asm
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/sync.o user/sync.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_sync user/sync.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_sync > user/sync.asm
riscv64-unknown-elf-objdump -t user/_sync | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sync.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/cowtest.o user/cowtest.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_cowtest user/cowtest.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_cowtest > user/cowtest.asm
riscv64-unknown-elf-objdump -t user/_cowtest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/cowtest.sym
mkfs/mkfs fs.img README  user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync  user/_cowtest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1017 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ cowtest
simple: ok
simple: ok
three: ok
three: ok
three: ok
file: ok
forkfork: ok
ALL COW TESTS PASSED
$ usertests -q
usertests starting
test copyin: OK
test copyout: OK
```

```
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/sysfile.o kernel/sysfile.c
riscv64-unknown-elf-gcc -march=rv64gc -g -ffile-prefix-map=/home/saad/xv6-labs-2026=. -c -o kernel/kernelvec.o kernel/kernelvec.S
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/plic.o kernel/plic.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/virtio_disk.o kernel/virtio_disk.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/start.o kernel/start.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/console.o kernel/console.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/printk.o kernel/printk.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/uart.o kernel/uart.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_COW -DLAB_COW -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o ker
nel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printk.o kernel
/uart.o kernel/spinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/saad/xv6-labs-2026'
== Test running cowtest ==
$ make qemu-gdb
(34.5s)
== Test   simple ==
  simple: OK
== Test   three ==
  three: OK
== Test   file ==
  file: OK
== Test   forkfork ==
  forkfork: OK
== Test usertests ==
$ make qemu-gdb
(75.1s)
== Test   usertests: copyin ==
  usertests: copyin: OK
== Test   usertests: copyout ==
  usertests: copyout: OK
== Test   usertests: all tests ==
  usertests: all tests: OK
== Test time ==
time: OK
Score: 130/130
saad@sand-dweller:~/xv6-labs-2026$
```